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Automated memory overclocking

Abstract

Automated memory overclocking is described. In accordance with the described techniques, one or more sets of overclocked memory settings of a memory are automatically selected for performance testing and stability testing of the memory. The one or more sets of the overclocked memory settings are tested for performance of the memory and a performance indication is output for each of the one or more sets of the overclocked memory settings. The one or more sets of the overclocked memory settings are tested for stability of the memory and a stability indication is output for each of the one or more sets of the overclocked memory settings. One of the one or more sets of the overclocked memory settings are selected as optimized overclocked memory settings for the memory.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6219241	12/2000	Jones	N/A	N/A
7382366	12/2007	Klock	714/E11.154	G06F 11/24
7469355	12/2007	Chong	N/A	N/A
7500122	12/2008	Won et al.	N/A	N/A
10852761	12/2019	Irshad et al.	N/A	N/A
11262924	12/2021	Alverson et al.	N/A	N/A
11609596	12/2022	Sin et al.	N/A	N/A
11977757	12/2023	Ley et al.	N/A	N/A
12038779	12/2023	Mehra et al.	N/A	N/A
2002/0161609	12/2001	Zizzamia	705/4	G06Q 40/08
2003/0065915	12/2002	Yu et al.	N/A	N/A
2003/0097602	12/2002	Bigbee et al.	N/A	N/A
2003/0202512	12/2002	Kennedy	370/395.31	H04W 40/14
2005/0266711	12/2004	Petersen	439/225	G11C 5/147
2006/0146638	12/2005	Chang	257/E27.099	G11C 11/413
2006/0149975	12/2005	Rotem et al.	N/A	N/A
2006/0280018	12/2005	Cases	365/226	G06F 13/4072
2007/0035985	12/2006	Fifield	365/154	G11C 11/413
2007/0044049	12/2006	Adams	700/109	G06F 30/367
2007/0106428	12/2006	Omizo et al.	N/A	N/A
2007/0183179	12/2006	Maeda	365/52	G06F 3/0659
2008/0114973	12/2007	Norton et al.	N/A	N/A
2008/0230893	12/2007	Too et al.	N/A	N/A
2008/0276026	12/2007	Branover et al.	N/A	N/A

2010/0131221	12/2009	Chien	702/81	G11C 29/023
2010/0220516	12/2009	Lee	365/158	G06F 30/39
2011/0016214	12/2010	Jackson	N/A	N/A
2011/0040902	12/2010	Housty	N/A	N/A
2011/0161706	12/2010	Huang et al.	N/A	N/A
2011/0231637	12/2010	Schuette	N/A	N/A
2014/0006864	12/2013	Menon	714/E11.178	G11C 29/022
2014/0032826	12/2013	Lee et al.	N/A	N/A
2014/0095904	12/2013	Ananthakrishnan et al.	N/A	N/A
2014/0229666	12/2013	Schoenborn	711/167	G11C 11/4074
2015/0016191	12/2014	Tsai et al.	N/A	N/A
2015/0270018	12/2014	Chinnakkonda Vidyapoornachary	714/721	G11C 29/50
2016/0084919	12/2015	Gokaraju	702/182	H02P 9/102
2017/0262354	12/2016	Han	N/A	N/A
2018/0095806	12/2017	Aneja et al.	N/A	N/A
2018/0188769	12/2017	Chen et al.	N/A	N/A
2018/0188770	12/2017	Chen	N/A	N/A
2019/0079806	12/2018	Ragland et al.	N/A	N/A
2020/0019462	12/2019	Prather et al.	N/A	N/A
2020/0049157	12/2019	Richter	N/A	N/A
2020/0097201	12/2019	Irshad et al.	N/A	N/A
2020/0192420	12/2019	Irshad	N/A	G09G 5/39
2020/0364041	12/2019	Bulusu et al.	N/A	N/A
2021/0068302	12/2020	North et al.	N/A	N/A
2021/0081127	12/2020	Xiao et al.	N/A	N/A
2021/0191450	12/2020	Mehra	N/A	G06F 1/3296
2021/0200456	12/2020	Alverson	N/A	G11C 29/10
2021/0326232	12/2020	Partiwala	N/A	G06F 11/3428
2021/0373642	12/2020	Wang et al.	N/A	N/A
2022/0093197	12/2021	Wu et al.	N/A	N/A
2022/0113757	12/2021	Yan	N/A	G06F 1/3243
2022/0113881	12/2021	Roberts	N/A	N/A
2022/0155982	12/2021	Alverson et al.	N/A	N/A
2022/0179706	12/2021	Khosrowpour	N/A	G06F 11/3466
2022/0197832	12/2021	Ruttenberg et al.	N/A	N/A
2023/0107735	12/2022	Wang	370/235	H04L 67/02

2023/0215475	12/2022	Chien	365/193	G11C 7/1021
2023/0315171	12/2022	Ahrens et al.	N/A	N/A
2023/0315191	12/2022	Alverson et al.	N/A	N/A
2023/0324967	12/2022	Ahrens et al.	N/A	N/A
2023/0350591	12/2022	Ley et al.	N/A	N/A
2023/0350696	12/2022	Harwani et al.	N/A	N/A
2023/0350715	12/2022	Knight et al.	N/A	N/A
2023/0409368	12/2022	Siddiqui	N/A	G06F 1/08
2024/0143445	12/2023	Leong et al.	N/A	N/A
2024/0211142	12/2023	Leong et al.	N/A	N/A
2025/0216888	12/2024	Rodrigue et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
2665787	12/2003	CN	N/A
208673319	12/2018	CN	N/A
2023211606	12/2022	WO	N/A
2024144868	12/2023	WO	N/A

OTHER PUBLICATIONS

U.S. Appl. No. 17/708,453 , “Final Office Action”, U.S. Appl. No. 17/708,453, filed Jul. 20, 2023, 12 pages. cited by applicant
U.S. Appl. No. 17/708,453 , “Non-Final Office Action”, U.S. Appl. No. 17/708,453, filed Mar. 14, 2023, 11 pages. cited by applicant
U.S. Appl. No. 17/708,453 , “Non-Final Office Action”, U.S. Appl. No. 17/708,453, filed Sep. 8, 2023, 13 pages. cited by applicant
U.S. Appl. No. 17/732,718 , “Non-Final Office Action”, U.S. Appl. No. 17/732,718, filed Sep. 14, 2023, 17 pages. cited by applicant
Cutress, Ian , “Intel Launches Xeon-W CPUs for Workstations: Skylake-SP & ECC for LGA2066”, AnandTech [Online][retrieved Oct. 11, 2023]. Retrieved from the Internet < https://www.anandtech.com/show/11775/intel-launches-xeon-w-cpus-for-workstations>, Aug. 29, 2017, 8 pages. cited by applicant
PCT/US2023/014176 , “International Search Report and Written Opinion”, PCT Application No. PCT/US2023/014176, Jun. 12, 2023, 8 pages. cited by applicant
PCT/US2023/062509 , “International Search Report and Written Opinion”, PCT Application No. PCT/US2023/062509, May 26, 2023, 10 pages. cited by applicant
PCT/US2023/063347 , “International Search Report and Written Opinion”, PCT Application No. PCT/US2023/063347, Jun. 13, 2023, 8 pages. cited by applicant
Rodrigue, Wayne Paul, et al., “Pursuant to MPEP § 2001.06(b) the applicant brings the following co-pending application to the Examiner's attention:”, U.S. Appl. No. 18/399,052, filed Dec. 28, 2023, 52 pages. cited by applicant
Knight, Joshua Taylor, et al., “US Application as Filed”, U.S. Appl. No. 17/732,987, filed Apr. 29, 2022, 41 pages. cited by applicant
Leong, Alicia Wen Ju Yurie, et al., “US Application as Filed”, U.S. Appl. No. 17/974,981, filed Oct. 27, 2022, 41 pages. cited by applicant
Leong, Alicia Wen Ju Yurie, et al., “US Application as Filed”, U.S. Appl. No. 18/146,538, filed Dec. 27, 2022, 46 pages. cited by applicant
U.S. Appl. No. 17/974,981 , “Non-Final Office Action”, U.S. Appl. No. 17/974,981, filed Jan. 31,

2024, 11 pages. cited by applicant
PCT/US2023/036338 , “International Seach Report and Written Opinion”, PCT Application No. PCT/US2023/036338, Feb. 20, 2024, 7 pages. cited by applicant
U.S. Appl. No. 17/974,981 , “Final Office Action”, U.S. Appl. No. 17/974,981, filed May 24, 2024, 13 pages. cited by applicant
“Non-Final Office Action”, U.S. Appl. No. 17/732,987, filed Nov. 22, 2024, 27 pages. cited by applicant
“Final Office Action”, U.S. Appl. No. 17/732,987, May 14, 2025, 17 pages. cited by applicant
“Non-Final Office Action”, U.S. Appl. No. 17/974,981, Mar. 10, 2025, 12 pages. cited by applicant
“Non-Final Office Action”, U.S. Appl. No. 18/399,052, Jun. 4, 2025, 15 pages. cited by applicant

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Background/Summary

BACKGROUND
(1) Memory, such as random access memory (RAM), stores data that is used by the processor of a computing device. Performance of a memory generally depends on its operating frequency as well as its latency characteristics, which are often referred to as “timings”. A memory can be “overclocked” by modifying specific parameters of the memory in order to achieve faster operating speeds which improves the performance of the computing device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a block diagram of a non-limiting example system having a memory and a memory controller operable to implement automated memory overclocking.
(2) FIG. **2** depicts a non-limiting example in which performance and stability of memory settings is tested as part of automating overclocking of memory.
(3) FIG. **3** depicts a non-limiting example of a user interface in one or more implementations.
(4) FIG. **4** depicts a non-limiting example of a user interface in one or more implementations.
(5) FIG. **5** depicts a non-limiting example of a user interface in one or more implementations.
(6) FIG. **6** depicts a procedure in an example implementation of automated memory overclocking.
DETAILED DESCRIPTION
Overview
(7) Conventional approaches for memory overclocking utilize target memory setting profiles which are often generated based on a “worst case” perspective. Even though these memory setting profiles may be above default memory capabilities, they are still quite conservative and are not customized based on the parameters of specific computing system. Users that wish to further optimize their memory above these default settings are often forced to rely on third party software that generates target setting based on past data. This past data, however, is based on capabilities of other computing systems and does not take into account the hardware capabilities of the specific computing system.
(8) To solve these problems, automated memory overclocking is described. The described techniques automate the process of determining optimized overclocking settings for a computing

system. To do so, the describes techniques test both performance and stability of memory settings for operating the memory, such as to test performance and stability of one or more sets of overclocked memory settings. For example, the techniques discussed herein allow various overclocked memory settings to be adjusted and a performance test run using the adjusted overclocked memory settings. In one or more implementations, a memory controller runs the performance test on the memory (e.g., by executing a workload on the memory), which generates performance values using the overclocked memory settings (or using adjusted overclocked memory settings). This process is automatically repeated multiple times for various memory settings and various overclocked values for those memory settings. The performance values generated by running the performance test are used to generate a performance indication that identifies how sensitive performance of the memory is to the one or more overclocked memory settings. The performance indication identifies which one or more overclocked memory settings have the most effect on operation of the memory in relation to a specified optimization that the user is interested in, allowing the user to then focus on simply selecting an optimization, such as increased performance of the memory, reduced power usage, reduced latency, and so forth. This frees a user from needing to guess or take a trial and error approach to determine which of the one or more memory settings to adjust to achieve the optimization that the user desires.

(9) The memory controller also tests a stability of the overclocked memory settings which, according to the one or more performance tests, affect the operation of memory for a specified optimization. This ensures that to achieve the specified optimization, the controller sets the overclocked memory settings set within a range of values that will keep the memory system stable (e.g., within a guardband of the memory's margins). This allows the one or more overclocked memory settings having the most effect on the desired optimization of memory operation system performance to be selected while reducing or eliminating the risk that the selected settings may render the memory system nonfunctional.

(10) In some aspects, the techniques described herein relate to a system including: a memory, a performance scoring component to test overclocked memory settings for the memory and output a performance indication for the overclocked memory settings, a stability testing component to test the overclocked memory settings and output a stability indication that predicts a stability of the memory over time if the memory is configured to operate with the overclocked memory settings, and a setting selector component configured to output optimized overclocked memory settings based on the performance indication output by the performance scoring component and the stability indication output by the stability testing component.

(11) In some aspects, the techniques described herein relate to a system, wherein the setting selector component is further configured to control testing by the performance scoring component and the stability testing component until the optimized overclocked memory settings are determined.

(12) In some aspects, the techniques described herein relate to a system, wherein the setting selector component is further configured to provide the overclocked memory settings to the performance scoring component and the stability testing component for testing.

(13) In some aspects, the techniques described herein relate to a system, wherein the setting selector component is further configured to provide additional overclocked memory settings to the performance scoring component and the stability testing component for testing based on at least one of the performance indication or a predicted stability of the memory.

(14) In some aspects, the techniques described herein relate to a system, wherein the setting selector component is further configured to provide the additional overclocked memory settings to the performance scoring component and the stability testing component for testing responsive to the a prediction indicating that the memory is too unstable if the memory is configured to operate with the overclocked settings, wherein the additional overclocked memory settings provided by the setting selector component are configured to less aggressively overclock the memory than the

overclocked memory settings.
(15) In some aspects, the techniques described herein relate to a system, wherein the setting selector component is further configured to provide the additional overclocked memory settings to the performance scoring component and the stability testing component for testing responsive to the a prediction indicating that the memory is stable if the memory is configured to operate with the overclocked settings and the predicted stability falling outside of a marginal stability guardband, wherein the additional overclocked memory settings provided by the setting selector component are configured to more aggressively overclock the memory than the overclocked memory settings.
(16) In some aspects, the techniques described herein relate to a system, wherein the performance scoring component and the stability testing component test the overclocked memory settings during a boot up process of the system.
(17) In some aspects, the techniques described herein relate to a system, wherein the setting selector is further configured to automatically configure the memory to operate with the optimized overclocked memory settings.
(18) In some aspects, the techniques described herein relate to a system, wherein the setting selector component is further configured to output the optimized overclocked memory settings in a user interface.
(19) In some aspects, the techniques described herein relate to a system, wherein the performance scoring component, the stability testing component, and the setting selector component are implemented in a memory controller.
(20) In some aspects, the techniques described herein relate to a method including: selecting one or more sets of overclocked memory settings of a memory for performance testing and stability testing of the memory, testing the one or more sets of the overclocked memory settings for performance of the memory and outputting a performance indication for each of the one or more sets of the overclocked memory settings, testing the one or more sets of the overclocked memory settings for stability of the memory and outputting a stability indication for each of the one or more sets of the overclocked memory settings, and selecting one of the one or more sets of the overclocked memory settings as optimized overclocked memory settings for the memory.
(21) In some aspects, the techniques described herein relate to a method, further including automatically configuring memory to operate with the optimized overclocked memory settings.
(22) In some aspects, the techniques described herein relate to a method, further including outputting the optimized overclocked memory settings in a user interface.
(23) In some aspects, the techniques described herein relate to a method, wherein the user interface includes one or more controls to enable adjustment of the optimized overclocked memory settings.
(24) In some aspects, the techniques described herein relate to a method, wherein the selecting one or more sets of overclocked memory settings of a memory for testing, the testing the one or more sets of the overclocked memory settings for performance of the memory and outputting a performance indication for each of the one or more sets of the overclocked memory settings, the testing the one or more sets of the overclocked memory settings for stability of the memory over time and outputting a stability indication for each of the one or more sets of the overclocked memory settings, and the selecting one of the one or more sets of the overclocked memory settings as optimized overclocked memory settings for the memory occur during a boot up process.
(25) In some aspects, the techniques described herein relate to a computing device including: a memory, and a memory controller to: initiate a boot up process of the computing device to automatically select optimized overclocked memory settings for the memory, and during the boot up process of the computing device: automatically identify one or more sets of overclocked memory settings for testing, test the one or more sets of the overclocked memory settings for performance of the memory and output a performance indication for each of the one or more sets of the overclocked memory settings, test the one or more sets of the overclocked memory settings for stability of the memory and output a stability indication for each of the one or more sets of the

overclocked memory settings, and select one of the one or more sets of the overclocked memory settings as the optimized overclocked memory settings.

(26) In some aspects, the techniques described herein relate to a computing device, wherein the memory controller is further configured to automatically configure the memory to operate with the optimized overclocked memory settings.

(27) In some aspects, the techniques described herein relate to a computing device, wherein the memory controller is further configured to output the optimized overclocked memory settings in a user interface.

(28) In some aspects, the techniques described herein relate to a computing device, wherein the user interface enables adjustment of the optimized overclocked memory settings.

(29) In some aspects, the techniques described herein relate to a computing device, wherein the memory controller initiates the boot up process responsive to user input to a user interface to select optimized overclocked memory settings.

(30) FIG. **1** is a block diagram of a non-limiting example system **100** having a memory and a memory controller operable to implement automated memory overclocking. In particular, the system **100** includes memory **102**, memory controller **104**, and processing unit **106**. In accordance with the described techniques, the memory **102**, the memory controller **104**, and the processing unit **106** are coupled to one another via one or more wired or wireless connections. Example wired connections include, but are not limited to, buses (e.g., a data bus), interconnects, through silicon vias, traces, and planes. Examples of a device or apparatus in which the system **100** is configured for integration include, but are not limited to, servers, personal computers, laptops, desktops, game consoles, set top boxes, tablets, smartphones, mobile devices, virtual and/or augmented reality devices, wearables, medical devices, systems on chips, and other computing devices or systems.

(31) The memory **102** is a device or system that is used to store information, such as for immediate use in a device. In one or more implementations, the memory **102** corresponds to semiconductor memory where data is stored within memory cells on one or more integrated circuits. In at least one example, the memory **102** corresponds to or includes volatile memory, examples of which include random-access memory (RAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM). Alternatively or in addition, the memory **102** corresponds to or includes non-volatile memory, examples of which include flash memory, read-only memory (ROM), programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electronically erasable programmable read-only memory (EEPROM), and non-volatile dual in-line memory module (DIMM) (NVDIMM).

(32) In one or more implementations, the memory **102** is configured as a dual in-line memory module (DIMM). A DIMM includes a series of dynamic random-access memory integrated circuits, and the modules are mounted on a printed circuit board. Examples of types of DIMMs include, but are not limited to, synchronous dynamic random-access memory (SDRAM), double data rate (DDR) SDRAM, double data rate 2 (DDR2) SDRAM, double data rate 3 (DDR3) SDRAM, double data rate 4 (DDR4) SDRAM, and double data rate 5 (DDR5) SDRAM. In at least one variation, the memory **102** is configured as a small outline DIMM (SO-DIMM) according to one of the above-mentioned SDRAM standards, e.g., DDR, DDR2, DDR3, DDR4, and DDR5. In one or more implementations, the memory **102** is low-power double data rate (LPDDR), also known as LPDDR SDRAM, and is a type of synchronous dynamic random-access memory. In variations, LPDDR consumes less power than other types of memory and/or has a form factor suitable for mobile computers and devices, such as mobile phones. Examples of LPDDR include, but are not limited to, low-power double data rate 2 (LPDDR2), low-power double data rate 3 (LPDDR3), low-power double data rate 4 (LPDDR4), and low-power double data rate 5 (LPDDR5). It is to be appreciated that the memory **102** is configurable in a variety of ways without departing from the spirit or scope of the described techniques.

(33) The memory controller **104** manages the memory **102**, including the communication of data to

and from the memory **102**. For example, the memory controller **104** manages the communication of data to the memory **102** from the processing unit **106** and the communication of data from the memory **102** to the processing unit **106**, e.g., over a coupling between the memory **102** and the processing unit **106**. As discussed above and below, the memory controller **104** also tests (e.g., during a boot up process) performance and stability of settings (e.g., clock and/or power settings) according to which the memory **102** is or can be configured to operate. In at least one variation, such settings are specified in one or more memory profiles.

(34) The processing unit **106** is a component that requests access to data from the memory **102** for performing one or more operations in relation to such data, e.g., in connection with executing an application (not shown). Examples of the processing unit include, but are not limited to, a central processing unit, a parallel accelerated processor (e.g., a graphics processing unit), a digital signal processor, a hardware accelerator, a microcontroller, a processing-in-memory (PIM) component, and a system on chip. In variations, different types of processing units are useable in the system in accordance with the described techniques.

(35) In the illustrated example, the memory controller **104** is depicted having an automated setting system **108** and memory control data **110**. The automated setting system **108** is depicted including various components, which are discussed in more detail below. It is also to be appreciated that the automated setting system **108** (and/or one or more of its components) is not included in the memory controller **104** and is implemented remotely from the memory controller **104**. The memory control data **110** includes one or more non-overclocking memory profiles **112** and one or more overclocking memory profiles **114**. The non-overclocking memory profiles **112** are implemented using memory settings **116** whereas the overclocking memory profiles **114** are implemented using overclocked memory settings **118**.

(36) In accordance with the described techniques, the overclocked memory settings **116** include at least one memory setting that is overclocked. In one or more scenarios, not all memory settings of an overclocking memory profile **114** are overclocked. In other scenarios, however, all the memory settings of an overclocking memory profile **114** are overclocked. In one or more implementations, a memory setting that is “overclocked” exceeds a certified value of the setting. For example, a clock rate set to exceed the clock rate certified by a manufacturer of the memory **102** is “overclocked.” By contrast, a memory setting that is not “overclocked” does not exceed the certified value of the setting, e.g., a clock rate that is not “overclocked” does not exceed the clock rate certified by a manufacturer of the memory **102**. Broadly, use of overclocked memory setting(s) **118** and/or use of an overclocking memory profile **114** enables the memory **102** to operate in an overclocking mode.

(37) In one or more implementations, the memory settings **116** and the overclocked memory settings **118** include various clock and/or power settings. Example settings include, but are not limited to, a data rate (e.g., megatransfers per second), a number of cycles between sending a column address to memory and the beginning of data in a response (e.g., CAS or tCAS), a minimum number of clock cycles to open a row and access a column (e.g., tRCD), a measure of latency between issuing a precharge command to idle or close open row and an activate command to open a different row (e.g., tRP), a minimum number of clock cycles between a row active command and issuing a precharge command (e.g., tRAS), an active to active delay time within the same bank group (e.g., tRRDL), an active to active delay time to a different bank group (e.g., tRRDS), a four active window time (e.g., tFAW), a write recovery time (e.g., tWR), a column address strobe (CAS) latency time (e.g., tAA), an active to active/refresh delay time (e.g., tRC), a refresh recovery delay time in a normal refresh mode (tRFC1), a refresh recovery delay time in a fine granularity refresh mode (tRFC2), a refresh recovery delay time in a same bank refresh mode (tRFCs), a write to read command delay second within a same bank group (tWTRL), a write to read command delay second within a different bank group (tWTRS), a read to precharge delay (tRTP), nominal power supply voltage (e.g., VDD), output stage drain power voltage (e.g., VDDQ), programming power voltage (e.g., VPP), and so forth. It is to be appreciated that the

memory settings **116** and/or the overclocked memory settings **118** specify values for one or more of those settings and/or various other settings associated with operating memory without departing from the spirit or scope of the described techniques.

(38) In accordance with the described techniques, the automated setting system **108** of the memory controller **104** and/or another component of the system **100** (e.g., the physical layer (PHY)) is configured to test performance and stability of memory settings for operating the memory **102**, such as such as to test performance and stability of the memory settings **116** of one or more non-overclocking memory profiles **112** and/or the overclocked memory settings **118** of one or more overclocking memory profiles **114**. For example, the automated setting system **108** tests performance of the memory settings to determine which memory settings, if adjusted, most affect operation of the memory **102** to achieve one or more optimization objectives, e.g., optimized performance of the memory **102**, reduced power usage by the memory **102**, reduced latency of the memory **102**, and so forth. The automated setting system **108** also tests a stability of memory settings, such by testing a stability of one or more overclocked values of the memory settings that have been identified to most affect the operation of the memory **102** in connection with achieving one or more specified optimization objectives. The automated setting system **108** tests the performance and stability of such memory settings during a boot up phase of the system **100**. In one or more implementations, the automated setting system **108** performs iterations of testing one or more sets of overclocked memory settings **118** for performance and testing one or more sets of overclocked memory settings **118** for stability. In at least one variation, all sets of overclocked memory settings that were performance tested are also stability tested. However, in at least one other variation, fewer than all the sets of overclocked memory settings that were performance tested are also stability tested. Rather than stability testing all the sets, for instance, a subset of sets of the overclocked memory settings that were performance tested are also stability tested, such as sets of the overclocked memory settings having performance values indicating that they sufficiently affect (e.g., above a threshold) operation of the memory to achieve one or more optimization objectives.

(39) In this example, the automated setting system **108** is depicted including a setting selector component **120**, a performance scoring component **122**, and a stability testing component **124**. In variations, these components are configured as firmware or intellectual property (IP) cores run on or otherwise accessible to the memory controller **104**. Alternatively, one or more of those components are implemented in hardware. Although depicted as part of the memory controller **104**, in one or more implementations, the automated setting system **108** and/or one or more of the setting selector component **120**, the performance scoring component **122**, and the stability testing component **124** are implemented remotely from the memory controller **104**. Additionally or alternatively, the automated setting system **108** includes more, fewer, or different components in variations to implement the described techniques.

(40) The illustrated example also includes an optimization request **126**. As discussed in more detail below, in one or more implementations, the optimization request **126** is received. For instance, the optimization request **126** is received responsive to user input selecting an option via a user interface to optimize the memory **102**. In at least one variation, such a user interface presents (e.g., displays) one or more optimization objectives from which a user can select for optimizing the memory **102**, examples of which include optimizing for performance of the memory **102** (e.g., increase a number read/write accesses handled, increase a frequency at which read/write accesses are handled, etc.), optimizing for power usage by the memory **102**, optimizing for latency of the memory **102**, and performing a default optimization in which one or more optimizations are weighted/combined to achieve an overall optimization, to name just a few.

(41) In one or more implementations, therefore, the optimization request **126** includes or otherwise indicates specified optimizations. Although it is discussed that user input triggers the optimization request **126**, in at least one variation, the optimization request **126** is triggered by an application,

such as when the application is launched, installed, or updated, for instance. Responsive to the optimization request **126**, the system **100** is powered down and then a boot up process is performed, during which overclocked memory settings are performance- and stability-tested (e.g., iteratively) by the memory controller **104** and/or one or more of its various components.
(42) In at least one variation, the setting selector component **120** selects one or more sets of the settings that are to be performance tested by the performance scoring component **122** and stability tested by the stability testing component **124**. By way of example, the setting selector component **120** selects at least a first set of overclocked memory settings **118** to be tested. In one or more scenarios, after this first set is tested, the setting selector component **120** selects at least a second set of overclocked memory settings **118** (e.g., a next set) to be tested based on a performance indication and stability indication generated in connection with testing the first set.
(43) In one or more implementations, the setting selector component **120** selects a next set of overclocked memory settings **118** to test until the performance indication and stability indication satisfy one or more thresholds and/or ranges. By way of example and not limitation, a next set of settings is selected and iterations of performance- and stability-testing are performed until the performance indication of a set of settings satisfies a threshold performance (or threshold improvement) and the stability indication of the set of settings falls within a marginal stability range (e.g., which indicates that operation of the memory **102** is “stable enough” if operated according to the set of settings). In one or more implementations, a marginal stability range is based on identifying values of settings that cause unstable operation of the memory **102** and buffering (e.g., with a guardband) those values so that the overclocked memory settings are not set to values within the buffer (or at determined unstable levels) causing unstable operation of the memory **102**. Further, the marginal stability range is within a threshold stability of the guardband. Thus, in one or more variations, the guardband defines a lowest level of stability for the marginal stability range, and the threshold stability added to the guardband defines a highest level of stability for the marginal stability range. In one or more implementations, the setting selector component **120** causes a next set of settings to be selected if the tested set is “not stable enough” (e.g., the stability indication reaches unstable levels or falls within the guardband of the unstable levels).
(44) In at least one variation, during the boot up process of the system **100**, the performance scoring component **122** tests performance of overclocked memory settings **118** for operating the memory **102** in accordance with the following discussion. The performance scoring component **122** runs the performance test with at least one set of the overclocked memory settings **118** and makes various read and write requests to the memory **102**. As noted above and below, in one or more variations, the performance test run by the performance scoring component **122** includes or otherwise corresponds to running one or more workloads with the memory **102**. For instance, different workloads are designed to test performance of the memory **102** in relation to different optimizations and generate performance values that identify settings which achieve (or improve performance toward) such optimizations, such as latency of the memory **102**, bandwidth of the memory **102**, power usage of the memory **102** (e.g., battery life for a notebook or other portable device, power for performance per watt calculations for a desktop computer), and so forth. When run on the memory **102**, a workload generates one or more performance values, which are provided to the performance scoring component **122** (or recorded by it).
(45) The performance scoring component **122** evaluates these performance values and generates a performance indication indicating how sensitive performance of the memory **102** is to the one or more memory settings tested. The performance indication is output to any of various different devices or entities. For instance, the performance indication is output to the setting selector component **120** and/or the stability testing component **124** to use the performance indication to determine automatically (e.g., without user input) which memory setting values to adjust in order to improve performance of memory **102** (e.g., overclock the memory **102** to achieve one or more optimization objectives). In at least one implementation, the performance indication is output via

display device (e.g., after the system **100** finishes the boot up process), which allows a user to see which memory settings affect (have the most effect and least effect on) achieving an optimization. Additionally or alternatively, an indication derived from the performance indication, that shows general sensitivity of the memory to one or more memory settings that were changed (e.g., a grade, a heatmap, and so forth as discussed in more detail below), is output to a display device or another application or system.

(46) In one or more implementations, the performance scoring component **122** adjusts one or more of multiple different memory settings multiple times, and the one or more workloads run during the performance test generate the one or more performance values for new overclocked memory settings **118** (or a set of them) after each adjustment. This allows the performance scoring component **122** to generate one or more performance indications identifying which overclocked memory settings **118** the memory **102** is most (and least) sensitive to for the workload(s) run during the performance test.

(47) It should be noted that although a single workload may be run as part of the performance testing performed by the performance scoring component **122**, in one or more implementations, the performance scoring component **122** runs more than one workload. In one or more implementations, the performance scoring component **122** generates one or more performance indications for each different workload. Additionally or alternatively, the performance scoring component **122** generates one or more generic or combined performance indications for the multiple different workloads.

(48) By determining through the performance testing which particular memory settings, if overclocked, affect the memory **102** (e.g., most or least) in terms of achieving an optimization objective, the stability testing can then be used to determine how much those memory settings can be adjusted (e.g., overclocked) and maintain operation of the memory **102** within a stable range, e.g., within a marginal stability.

(49) In at least one variation, the stability testing component **124** tests a stability of a set of the overclocked memory settings **118** for operating the memory **102** (e.g., an overclocking memory profile **114**) during a boot up process of the system **100**. For example, the setting selector component **120** specifies a set of overclocked memory settings **118** to stability test based on previous performance testing. For instance, the setting selector component **120** specifies the set of settings during the boot up process, e.g., during which the set of settings were previously performance tested by the performance scoring component **122**. During the boot up process of the system **100**, the stability testing component **124** of the memory controller **104** also tests the stability of the overclocked memory settings **118**. In at least one variation, the stability testing component **124** performs one or more initial stability tests on the set of overclocked memory settings **118**, and one or more additional stability tests if the memory settings “pass” the initial stability tests, e.g., the settings are indicated stable or at least marginally stable via the initial stability tests.

(50) After performing one or more stability tests on the memory **102**, one or more stability indications are returned, e.g., to the setting selector component **120**. Using the one or more stability indications, the setting selector component **120** determines whether to conduct a subsequent iteration of performance- and stability-testing or proceed to operating the memory **102** using the tested settings. In the scenario where the setting selector component **120** determines to proceed by operating the memory **102** using the tested memory settings, optimization results **128** are returned which indicate the selected set of overclocked memory settings **118**, e.g., optimized overclocked memory settings. In at least one variation, the memory controller **104** causes one or more change signals to be sent to change operation of the memory **102** so that it operates according to the overclocked memory settings **118** included in the optimization results **128**. In this way, the memory controller **104** automatically tests various combinations of overclocked memory settings and automatically selects a set of the overclocked memory settings for operating the memory **102** to

optimize performance of the memory **102** for achieving at least one optimization objective. As used herein "automatically test" refers to performance- and stability-testing various combinations of settings and their values without receiving user input to select which particular memory settings to test or at what values to test them, and "automatically" select refers to selecting values at which to set the overclocked memory settings without receiving user input specifying how to specifically set them.
(51) In one or more implementations, human-understandable information indicative of the optimization results **128** (e.g., one or more descriptions of the settings adjusted and their adjusted values) is presented (e.g., displayed) via a user interface. Generally, results of the one or more stability tests predict a stability of the memory **102** over a subsequent time period if the memory **102** is configured to operate with the set of overclocked memory settings. Alternatively or in addition, the memory controller **104** is configured to test stability of the settings for operating the memory **102** during a different phase from the boot-up phase, such as while the system **100** is in a "sleeping" state.
(52) In one or more implementations, the stability testing component **124** performs one or more stability tests of the memory **102** during the boot up phase—it is to be appreciated that the stability tests described above and below are performable by the stability testing component **124** and/or another component in variations, even if described herein as being performed by the memory controller **104**. The performed tests are configured to test a stability of at least one set of memory settings (e.g., a set of overclocked memory settings selected or adjusted by the setting selector component **120**), such as a stability of the memory **102** to operate using the at least one set of memory settings. By way of example, during training of the memory **102**, the stability testing component **124** determines stable margins for bitlines of the memory **102** to operate and uses those margins to determine optimal settings for initializing the hardware. In accordance with the described techniques, the stability testing component **124** also uses the stable margins to determine stability of the memory **102** for a subsequent period of time, e.g., hardware stability in the long term.
(53) Examples of the one or more tests the stability testing component **124** and/or other component perform on the memory **102** during boot up to test the stability of the memory **102** under various settings include, but are not limited to, performing reference voltage (Vref) and/or delay-locked loop (DLL) shmoos (e.g., to generate one or more "eye diagrams" of the memory **102** or portions of the memory **102**), performing virtual timing and signal analysis (vTSA), adjusting different voltage and/or delay levels to generate visualization of memory data eyes for different bits of the memory, checking one or more portions of the memory **102** for rowhammering, and performing one or more RSC tests. In one or more implementations, the stability testing component **124** generates one or more eye diagrams of the memory **102** using a combination (e.g., a multidimensional combination) of reference voltage (Vref) and DLL shmoos. In at least one variation, the reference voltage (Vref) is shmooed to determine if the signal is a '1' or a '0' in connection with creating the memory eye diagrams. Based on the generated eye diagrams, for instance, the stability testing component **124** obtains the margins at which the memory **102** is capable of performing given variations in the environment in which the memory **102** operates. In variations, different tests are performed to test the stability of the memory **102** without departing from the described techniques.
(54) By way of contrast to the described stability testing techniques, conventional approaches involve running one or more "stress tests" using software, such as via an operating system. Such tests can take several hours (up to days) to determine a stability of the system. This is at least in part because the conventional tests are performed via software, which involves access (e.g., by an operating system) via one or more cores (of a processing unit), then access via data fabric of the system, and then access to the memory being stress tested—a slower path than a direct path to memory. Additionally, conventional stress tests are limited to determining the stability of the

system for the environmental variables and workload at a time those tests are performed, but the tests do not guarantee that the system will perform stably in the long term, especially if operation of the memory approaches its margins.
(55) In accordance with the described techniques, the stability testing component **124** (e.g., as part of the memory controller **104**) accesses the memory **102** directly. In other words, the stability tests are run from the memory controller **104** directly into the memory **102**. As a result, the stability tests performed on the memory **102** to determine a stability of the overclocked memory settings **118** are faster than conventional stress tests. In order to perform one or more of the tests during boot up, the stability testing component **124** bypasses refreshes and other periodic background activities, which is unable to be performed in various scenarios where the system has already been booted up (e.g., when running software). Additionally, by performing the performance—and stability—tests during boot up by directly accessing the memory **102**, the performance scoring component **122** and the stability testing component **124** avoid background latencies that occur with operating systems and/or other software after the system has booted up.
(56) The illustrated example also depicts clock and power **130** inputs to the memory **102** and the memory controller **104**. In accordance with the described techniques, the memory controller **104** and/or another component is configured to set the clock and power **130** inputs to cause the memory **102** to operate according to a set of memory settings, such as a set of overclocked memory settings **118** indicated in the optimization results **128** or of an overclocked memory profile **114**.
(57) For example, the memory controller **104** causes the memory **102** to operate according to a set of the overclocked memory settings **118** by sending one or more change signals to a voltage generator (not shown) to adjust a supply voltage (e.g., VDD), such that the clock and power **130** inputs subsequently include the supply voltage as adjusted according to the change signals. Additionally or alternatively, the memory controller **104** sends a change signal to a reference clock generator (not shown) to change a frequency of a clock rate, such that the clock and power **130** inputs subsequently include a reference clock signal as adjusted according to the change signals. The memory controller **104** is operable to adjust the clock and power **130** inputs in various ways to produce the settings specified, e.g., in the optimization results **128**, for operating the memory **102**.
(58) FIG. **2** depicts a non-limiting example **200** in which performance and stability of memory settings is tested as part of automating overclocking of memory. The example **200** includes from FIG. **1** the memory **102**, the memory controller **104**, and the processing unit **106**. Further, the memory controller **104** is depicted including the setting selector component **120**, the performance scoring component **122**, and the stability testing component **124**. Although depicted as including these components, it is to be appreciated that in one or more implementations the memory controller **104** (or some other component) is considered to perform the operations described herein as being performed by the setting selector component **120**, the performance scoring component **122**, and/or the stability testing component **124**.
(59) The example **200** includes a variety of example communications and operations between the memory controller **104**, the memory **102**, and the processing unit **106** over time. In this example **200**, the communications and operations are positioned vertically based on time, such that communications and operations closer to a top of the example occur prior to communications or operations further from the top of the example. It follows also that communications or operations closer to a bottom of the example occur subsequent to communications or operations further from the bottom. The example **200** also depicts various phases and/or states of the system **100** or portions of the system **100**. These phases and/or states are also positioned in the example **200** vertically based on time, such that phases or states closer to a top of the example occur prior to phases, states, or communications further from the top.
(60) Here, the illustrated example **200** depicts the memory controller **104** receiving the optimization request **126**. In one or more implementations, the optimization request **126** is received based on user input, such as user input received via a displayed control (e.g., a button) to optimize

operation of the memory **102**. As discussed above and below, in variations, the described system optimizes operation of the memory **102** to achieve an optimization objective, such as optimized performance of the memory **102**, reduced power usage by the memory **102**, reduced latency of the memory **102**, and so forth.

(61) The illustrated example **200** also depicts a boot up phase **202** of the system **100**. In one or more implementations, the system **100** enters the boot up phase **202** based on receipt of the optimization request **126**. Although a powered off phase is not depicted, in one or more such implementations, the system **100** is powered off based on the optimization request **126**, and then the system **100** is triggered to boot up. During a powered off phase, the system **100** (and the memory **102**) is powered off, examples of which include a soft off (e.g., G2/S5 state as specified by Advanced Configuration and Power Interface (ACPI)) and a mechanical off (e.g., G3 state as specified by ACPI), which require a reboot to return to a working state (e.g., G0/S0 state as specified by ACPI). During the boot up phase **202**, the system **100** performs various operations, such as hardware initializations, to advance the system **100** to a working state.

(62) In accordance with the described techniques, the boot up phase **202** includes performance testing **204** and stability testing **206** of the memory **102**. Broadly, the setting selector component **120** controls testing by the performance scoring component **122** and the stability testing component **124** until optimized overclocked memory settings are determined. As part of this, for instance, the memory controller **104** (e.g., the setting selector component **120**) selects first overclocked memory settings **208** for performance testing **204** and stability testing **206**. The memory controller **104** (e.g., the performance scoring component **122**) tests a performance of the first overclocked memory settings **208** for operating the memory **102**. In one or more implementations, the performance scoring component **122** performs the performance testing **204**, at least in part, by running one or more workloads with the memory **102** that involve various read and/or write accesses to the memory **102**.

(63) As the performance scoring component **122** performs the performance testing **204** on the memory **102**, one or more workloads that are run as part of the testing generate performance values **210**, which describe performance of the memory **102** in terms of completing one or more memory-based tasks or operations, when configured according to the first overclocked memory settings **208**. The performance values **210** are then used individually or in combination to produce one or more performance indications for the first overclocked memory settings **208**.

(64) Additionally, the memory controller **104** (e.g., the stability testing component **124**) tests a stability of the first overclocked memory settings **208** for operating the memory **102**. As part of the stability testing **206**, in one or more implementations, the stability testing component **124** tests a stability of operating the memory **102** with the first overclocked memory settings **208** by performing reference voltage (Vref) and/or delay-locked loop (DLL) shmoos and/or by generating one or more “eye diagrams” of the memory **102** or portions of the memory **102**. Based on the stability testing **206**, for instance, the memory controller **104** obtains the margins at which the memory **102** is capable of performing.

(65) As the stability testing component **124** performs the stability testing **206** on the memory **102**, the memory controller **104** (e.g., the setting selector component **120** and/or the stability testing component **124**) receives stability notifications **212** (e.g., signals), which indicate a response of the memory **102** to the stability tests performed during the boot up phase **202**. For instance, as the stability testing component **124** performs reference voltage (Vref) and/or DLL shmoos in the memory **102**, the memory controller **104** receives stability notifications **212** indicative of operation of the memory **102** and/or margins of its operation, under various combinations of voltage and/or delay levels.

(66) In one or more scenarios, the setting selector component **120** causes a next set of overclocked memory settings **118** to be tested, e.g., with the performance testing **204** and the stability testing **206**. As mentioned above and below, for instance, the setting selector component **120** controls

testing by the performance scoring component **122** and the stability testing component **124** until optimized overclocked memory settings are determined. By way of example, the setting selector component **120** selects an additional set of overclocked memory settings that less aggressively overclock the memory **102** than the first overclocked memory settings **208**, such as in a scenario where the stability notifications **212** are processed to generate a prediction that the memory **102** is too unstable if configured to operate with the first overclocked memory settings **208**. Alternatively, the setting selector component **120** selects an additional set of overclocked memory settings that more aggressively overclock the memory **102** than the first overclocked memory settings **208**, such as in a scenario where the stability notifications **212** are processed to generate a prediction that the memory **102** is stable if configured to operate with the first overclocked memory settings **208** and the predicted stability falls outside of a marginal stability range (i.e., the first overclocked memory settings **208** are "too stable"). In other words, the setting selector component **120** performs a setting adjustment **214** on the first overclocked memory settings **208**, adjusting them to produce second overclocked memory settings **216**, which are different from the first overclocked memory settings **208**.

(67) The memory controller **104** (e.g., the setting selector component **120**) then provides the second overclocked memory settings **216** for performance testing **204** and stability testing **206**. The memory controller **104** (e.g., the performance scoring component **122**) tests a performance of the second overclocked memory settings **216** for operating the memory **102**. By way of example, the performance scoring component **122** performs the performance testing **204** on the second overclocked memory settings **216**, at least in part, by running one or more workloads with the memory **102** that involve various read and/or write accesses to the memory **102**. As the performance scoring component **122** performs the performance testing **204** using the second overclocked memory settings **216**, one or more workloads that are run as part of the testing generate performance values **218**, which describe performance of the memory **102** in terms of completing one or more memory-based tasks or operations when configured according to the second overclocked memory settings **216**. The performance values **218** are then used individually or in combination to produce one or more performance indications for the second overclocked memory settings **216**.

(68) Further, the memory controller **104** (e.g., the stability testing component **124**) tests a stability of the second overclocked memory settings **216** for operating the memory **102**. As part of the stability testing **206**, in one or more implementations, the stability testing component **124** tests a stability of operating the memory **102** with the second overclocked memory settings **216** by performing reference voltage (Vref) and/or delay-locked loop (DLL) shmoos and/or by generating one or more "eye diagrams" of the memory **102** or portions of the memory **102**. Based on the stability testing **206**, the memory controller **104** obtains the margins at which the memory **102** is capable of performing, when configured according to the second overclocked memory settings **216**.

(69) As the stability testing component **124** performs the stability testing **206** on the memory **102** for the second overclocked memory settings **216**, the memory controller **104** (e.g., the setting selector component **120** and/or the stability testing component **124**) receives stability notifications **220** (e.g., signals), which indicate a response of the memory **102** to the stability tests performed during the boot up phase **202**, e.g., if the memory **102** is configured according to the second overclocked memory settings **216**. For instance, as the stability testing component **124** performs reference voltage (Vref) and/or DLL shmoos in the memory **102**, the memory controller **104** receives stability notifications **220** indicative of operation of the memory **102** with the second overclocked memory settings **216** and/or margins of its operation, under various combinations of voltage and/or delay levels.

(70) The illustrated example **200** depicts ellipses in the boot up phase **202** to indicate that in variations the setting selector component **120** performs more than two iterations of selecting a set of settings, performance testing **204** the set of settings, stability testing **206** the set of settings, and

adjusting those settings to produce a next set of settings for testing. As noted above, for instance, the setting selector component **120** controls testing by the performance scoring component **122** and the stability testing component **124** until optimized overclocked memory settings are determined.
(71) Responsive to determination of optimized overclocked memory settings, the optimization results **128** are output. In one or more implementations, the optimization results **128** include or otherwise indicate the optimized overclocked memory settings. In at least one variation, the memory controller **104** (e.g., the setting selector component **120**) outputs the optimization results **128**, which causes the memory **102** to operate according to the optimized overclocked memory settings. For example, the memory controller **104** causes the memory **102** to operate according to the optimized overclocked memory settings by sending one or more change signals to a voltage generator (not shown) to adjust a supply voltage (e.g., VDD), such that the clock and power **130** inputs subsequently include the supply voltage as adjusted according to the change signals. Additionally or alternatively, the memory controller **104** sends a change signal to a reference clock generator (not shown) to change a frequency of a clock rate, such that the clock and power **130** inputs subsequently include a reference clock signal as adjusted according to the change signals.
(72) Additionally or alternatively, the memory controller **104** outputs the optimization results **128** by communicating them to software (e.g., via a handshake), such as to an operating system (not shown). Alternatively or additionally, an operating system further provides the test results to an application running on the operating system, such as to a memory management application. In accordance with the described techniques, the optimization results **128** are output (e.g., displayed) in a human-understandable format via a user interface, such as via a user interface of an operating system and/or a user interface of a memory management application. Examples of formats in which the optimization results **128** are output include, but are not limited to, numerical scores, stop-light icons, text labels describing hierarchical levels of optimization, graphical icons, graphs or charts depicting a performance and/or stability of the memory over time if the memory is configured to operate with the optimized overclocked memory settings, and one or more visualizations of generated eye diagrams, to name just a few. In variations, the optimization results **128** are output in different human-understandable ways indicative of performance and/or stability of the memory **102** under tested settings without departing from the spirit or scope of the described techniques.
(73) Also in this example **200**, the memory controller **104** provides one or more set optimized overclocked memory settings **222** signals to the memory **102** to cause the memory **102** to operate according to an optimized overclocked memory mode **224**. By selecting optimized overclocked memory settings through iterations of the performance testing **204** and the stability testing **206**, the described techniques automate memory overclocking so that memory overclocking can be performed without user input specifying values for overclocking one or more memory settings. In accordance with the described techniques, the memory controller **104** causes the memory **102** to operate according to the optimized overclocked memory mode **224**, such as subsequent to having operated according to a different mode, e.g., a non-optimized mode, an overclocked memory mode optimized for a different optimization objective (e.g., performance versus power usage), or an overclocked memory mode where one or more of the settings were overclocked based on user input, to name just a few.
(74) Additionally or alternatively, the optimization results **128** can include or be used to generate a recommendation for output. In one variation, for instance, such a recommendation recommends that a user select an optimization objective or combination of optimization objectives and allow the system **100** to automatically select overclocked memory settings to achieve those objectives. By way of example, a software application (e.g., a memory management application) creates and causes output (e.g., display) of a notification after a threshold amount of time (e.g., a number of hours, days, months, etc.) reminding a user of the system **100** to select an optimization objective and allow the system to automatically optimize overclocked memory settings to achieve the optimization.

(75) In this example **200**, the memory **102** is also depicted receiving data requests **226** originating from the processing unit **106** and providing data **228** to the processing unit **106** for servicing those requests, in accordance with the described techniques. In particular, the memory controller **104** receives one or more data requests **226** from the processing unit **106** and the memory controller **104** schedules the data request **226** to the memory **102**. The memory controller **104** then accesses (e.g., reads from) a portion of the memory **102** corresponding to the data request **226** (e.g., to retrieve respective data **228**). The memory controller **104** then provides the respective data **228** to the processing unit **106**. This process of requesting data, retrieving data from the memory **102** (e.g., reading the data from the memory **102**), and providing the data read back to the processing unit **106**, is performed while the memory **102** operates according to the optimized overclocked memory mode **224**. In accordance with the described techniques, the processing unit **106** can also cause the memory controller **104** to write data to the memory **102** while operating in the optimized overclocked memory mode **224**.
(76) Notably, by performing the stability testing on the automated overclocking settings, the memory controller **104** is able to determine the stability expectations of the system. Thus, in one or more implementations, after the automated memory overclocking is performed, the memory controller **104** can inform the user after a certain time period that the stability tests should be run again to ensure that the automated memory overclocking settings are still stable.
(77) In the context of user interfaces for automated memory overclocking, consider the following discussion of FIGS. **3-5**.
(78) FIG. **3** depicts a non-limiting example **300** of a user interface in one or more implementations. The example **300** includes a display device **302** outputting a memory overclocking user interface **304**.
(79) In the illustrated example **300**, the user interface **304** is depicted displaying a set of memory settings **306** according to which the memory **102** is operable. In this example **300**, the user interface **304** includes a control **308**, which is selectable (e.g., via user input) to initiate automated overclocking of the memory **102**. For instance, receipt of user input in relation to the control **308** corresponds to or triggers the optimization request **126**, which causes the system **100** to automatically select optimized overclocked memory settings for operating the memory **102**. This includes performing the performance testing **204** and the stability testing **206** for one or more sets of settings selected by the setting selector component **120** during the boot up phase **202** until optimized overclocked memory settings are determined.
(80) In one or more implementations, the user interface **304** allows a user to provide user input to adjust one or more settings according to which the memory **102** operates. Additionally or alternatively, the user interface **304** includes controls which allows a user to select to initiate performance testing **204** of such user selected settings and/or stability testing **206** of the user selected settings. Indeed, it is to be appreciated that in variations, user interfaces for automated memory overclocking are configured in different ways without departing from the spirit or scope of the described techniques.
(81) FIG. **4** depicts a non-limiting example **400** of a user interface in one or more implementations. The example **400** includes a display device **402** outputting a memory overclocking user interface **404**.
(82) Like the illustrated example **300**, the illustrated example **400** depicts the user interface **404** displaying a set of memory settings **406** according to which the memory **102** is operable. Also like the example **300**, the user interface **404** in the example **400** includes a control **408**, which is selectable (e.g., via user input) to initiate automated overclocking of the memory **102**. In contrast to the user interface **304**, however, the user interface **404** further includes an optimization control **410** for selecting one or more optimization objectives. For instance, receipt of user input in relation to the control **410** specifies one or more optimization objectives in one or more scenarios. Examples of such optimizations are described in more detail above. In one or more implementations,

specification of one or more such optimizations causes the optimization request **126** to include such optimizations so that the memory controller **104** determines optimized overclocked memory settings based on the specified optimizations included in the optimization request **126**. In one or more implementations, the user interface **404** includes additional controls which enable the user to select specific settings for automated memory overclocking. For example, the user interface **404** may include a control which enables navigation to an additional customization menu whereby users can specify exact setting to automatically overclock.

(83) Thus, in one example, receipt of user input in relation to the control **408** triggers generation of the optimization request **126** so that the optimization request **126** is generated to include one or more specified optimization objectives, e.g., objectives specified based on user input via the optimization control **410**. In accordance with the described techniques, the optimization request **126** then causes the system **100** to automatically select optimized overclocked memory settings for operating the memory **102**. This includes performing the performance testing **204** and the stability testing **206** for one or more sets of settings selected by the setting selector component **120** during the boot up phase **202** until optimized overclocked memory settings are determined.

(84) FIG. **5** depicts a non-limiting example **500** of a user interface in one or more implementations. The example **500** includes a display device **502** outputting a memory overclocking user interface **504**.

(85) In the illustrated example **500**, the user interface **504** is depicted displaying a notification **506** indicating that optimized overclocked memory settings have been determined (e.g., by the setting selector component **120**, the performance scoring component **122**, and the stability testing component **124** during the boot up phase **202**), and the memory **102** has been adjusted to operate according to the optimized overclocked memory settings. In one or more implementations, such a notification is displayed after the boot up phase **202** of the system and when the system **100** returns to a working state (e.g., G0/S0 state as specified by ACPI). As discussed above, for instance, the optimization results **128** are provided to an application (e.g., a memory management application) in one or more variations, and information in the optimization results **128** enables the application to display the notification **506** via its user interface **504**. It is to be appreciated that the notification **506** is just one example and, in variations, different information is displayable after optimized overclocked memory settings are determined by the system **100** automatically and the system returns from the boot up phase **202** to a working state. Since the optimized overclocked memory settings are determined automatically and without user input to select the values, such a notification is effective to inform a user of the system **100** regarding how the hardware of their system has been adjusted.

(86) Having discussed example systems and user interfaces for automated memory overclocking, consider the following example procedures.

(87) FIG. **6** depicts a procedure in an example **600** implementation of automated memory overclocking.

(88) One or more sets of overclocked memory settings of a memory are selected for performance testing and stability testing of the memory (block **602**). By way of example, the memory controller **104** (e.g., the setting selector component **120**) selects first overclocked memory settings **208** and/or second overclocked memory settings **216** for performance testing **204** and stability testing **206**. In some cases, this selection is automatic and may occur during a boot up phase **202** of the system **100**. In one or more implementations, the system **100** enters the boot up phase **202** based on receipt of the optimization request **126**.

(89) The one or more sets of the overclocked memory settings are tested for performance of the memory and a performance indication is output for each of the one or more sets of the overclocked memory settings (block **604**). By way of example, the memory controller **104** (e.g., the performance scoring component **122**) tests a performance of the first overclocked memory settings **208** and/or the second overclocked memory settings **216** for operating the memory **102**. In one or

more implementations, the performance scoring component **122** performs the performance testing **204**, at least in part, by running one or more workloads with the memory **102** that involve various read and/or write accesses to the memory **102**.

(90) As the performance scoring component **122** performs the performance testing **204** on the memory **102**, one or more workloads that are run as part of the testing generate performance values **210** and/or the performance values **218**, which describe performance of the memory **102** in terms of completing one or more memory-based tasks or operations, when configured according to the first overclocked memory settings **208** and/or the second overclocked memory settings **216**, respectively. The performance values **210** and/or the performance values **218** are then used individually or in combination to produce one or more performance indications for the first overclocked memory settings **208** and/or the second overclocked memory settings **216**.

(91) The one or more sets of the overclocked memory settings are tested for stability of the memory and a stability indication is output for each of the one or more sets of the overclocked memory settings (block **606**). By way of example, the memory controller **104** (e.g., the stability testing component **124**) tests a stability of the first overclocked memory settings **208** and/or the second overclocked memory settings **216** for operating the memory **102**. As part of the stability testing **206**, in one or more implementations, the stability testing component **124** tests a stability of operating the memory **102** with the first overclocked memory settings **208** and/or the second overclocked memory settings **216** by performing reference voltage (Vref) and/or delay-locked loop (DLL) shmoos and/or by generating one or more "eye diagrams" of the memory **102** or portions of the memory **102**. Based on the stability testing **206**, for instance, the memory controller **104** obtains the margins at which the memory **102** is capable of performing.

(92) As the stability testing component **124** performs the stability testing **206** on the memory **102**, the memory controller **104** (e.g., the setting selector component **120** and/or the stability testing component **124**) receives stability notifications **212** and/or stability notifications **220**, which indicate a response of the memory **102** to the stability tests performed during the boot up phase **202**. For instance, as the stability testing component **124** performs reference voltage (Vref) and/or DLL shmoos in the memory **102**, the memory controller **104** receives stability notifications **212** and/or stability notifications **220** indicative of operation of the memory **102** and/or margins of its operation, under various combinations of voltage and/or delay levels.

(93) Notably, the setting selector component **120** causes a next set of overclocked memory settings **118** to be tested, e.g., with the performance testing **204** and the stability testing **206**. For example, the setting selector component **120** controls testing by the performance scoring component **122** and the stability testing component **124** until optimized overclocked memory settings are determined. By way of example, the setting selector component **120** may first test the first overclocked memory settings **208** and then select an additional set of overclocked memory settings that less aggressively overclock the memory **102** than the first overclocked memory settings **208**, such as in a scenario where the stability notifications **212** are processed to generate a prediction that the memory **102** is too unstable if configured to operate with the first overclocked memory settings **208**. Alternatively, the setting selector component **120** selects an additional set of overclocked memory settings that more aggressively overclock the memory **102** than the first overclocked memory settings **208**, such as in a scenario where the stability notifications **212** are processed to generate a prediction that the memory **102** is stable if configured to operate with the first overclocked memory settings **208** and the predicted stability falls outside of a marginal stability range (i.e., the first overclocked memory settings **208** are "too stable").

(94) One of the one or more sets of the overclocked memory settings are selected as optimized overclocked memory settings for the memory (block **608**). By way of example, the setting selector component **120** controls testing by the performance scoring component **122** and the stability testing component **124** until optimized overclocked memory settings are determined. Responsive to determination of optimized overclocked memory settings, the optimization results **128** are output.

In one or more implementations, the optimization results **128** include or otherwise indicate the optimized overclocked memory settings. In at least one variation, the memory controller **104** (e.g., the setting selector component **120**) outputs the optimization results **128**, which causes the memory **102** to operate according to the optimized overclocked memory settings. For example, the memory controller **104** causes the memory **102** to operate according to the optimized overclocked memory settings by sending one or more change signals to a voltage generator (not shown) to adjust a supply voltage (e.g., VDD), such that the clock and power **130** inputs subsequently include the supply voltage as adjusted according to the change signals. Additionally or alternatively, the memory controller **104** sends a change signal to a reference clock generator (not shown) to change a frequency of a clock rate, such that the clock and power **130** inputs subsequently include a reference clock signal as adjusted according to the change signals.
(95) It should be understood that many variations are possible based on the disclosure herein. Although features and elements are described above in particular combinations, each feature or element is usable alone without the other features and elements or in various combinations with or without other features and elements.
(96) The various functional units illustrated in the figures and/or described herein (including, where appropriate, the memory **102**, the memory controller **104**, the processing unit **106**, the setting selector component **120**, the performance scoring component **122**, and the stability testing component **124**) are implemented in any of a variety of different manners such as hardware circuitry, software or firmware executing on a programmable processor, or any combination of two or more of hardware, software, and firmware. The methods provided are implemented in any of a variety of devices, such as a general purpose computer, a processor, or a processor core. Suitable processors include, by way of example, a general purpose processor, a special purpose processor, a conventional processor, a digital signal processor (DSP), a graphics processing unit (GPU), a parallel accelerated processor, a plurality of microprocessors, one or more microprocessors in association with a DSP core, a controller, a microcontroller, Application Specific Integrated Circuits (ASICs), Field Programmable Gate Arrays (FPGAs) circuits, any other type of integrated circuit (IC), and/or a state machine.
(97) In one or more implementations, the methods and procedures provided herein are implemented in a computer program, software, or firmware incorporated in a non-transitory computer-readable storage medium for execution by a general purpose computer or a processor. Examples of non-transitory computer-readable storage mediums include a read only memory (ROM), a random access memory (RAM), a register, cache memory, semiconductor memory devices, magnetic media such as internal hard disks and removable disks, magneto-optical media, and optical media such as CD-ROM disks, and digital versatile disks (DVDs).

Claims

1. A system comprising: a memory; and a memory controller to: iteratively test performance and stability of overclocked memory settings to determine optimized overclocked memory settings by: testing performance of a first set of the overclocked memory settings; testing stability of the first set of overclocked memory settings; testing performance of at least a second set of the overclocked memory settings; and testing stability of the second set of the overclocked memory settings; and output the optimized overclocked memory settings.
2. The system of claim 1, wherein the memory controller is further configured to control testing performance and stability until the optimized overclocked memory settings are determined.
3. The system of claim 1, wherein the memory controller is configured to iteratively test at least the first set and second set of the overclocked memory settings during a boot up process of the system.
4. The system of claim 1, wherein the memory controller is further configured to automatically configure the memory to operate with the optimized overclocked memory settings.

5. The system of claim 1, wherein the memory controller is further configured to output the optimized overclocked memory settings in a user interface.
6. The system of claim 1, wherein the second set of overclocked memory settings is selected based on a predicted stability of the first set of overclocked memory settings determined from the testing stability of the first set of overclocked memory settings.
7. The system of claim 6, wherein the second set of overclocked memory settings is selected to more aggressively overclock the memory compared to the first set of overclocked memory settings when the memory is predicted to be stable if configured to operate with the first set of overclocked memory settings.
8. The system of claim 6, wherein the second set of overclocked memory settings is selected to less aggressively overclock the memory compared to the first set of overclocked memory settings when the memory is predicted to be unstable if configured to operate with the first set of overclocked memory settings.
9. The system of claim 6, wherein the memory controller is further configured to determine whether to conduct a subsequent iteration of testing performance and stability for at least a third set of overclocked memory settings based on a predicted stability of the second set of overclocked memory settings determined from the testing stability of the second set of overclocked memory settings.
10. The system of claim 1, wherein the memory controller is further configured to select the first set or the second set of the overclocked memory settings as the optimized overclocked memory settings.
11. The system of claim 1, wherein the first set of the overclocked memory settings are different than the second set of overclocked memory settings.
12. A method comprising: selecting a first set of overclocked memory settings of a memory for performance testing and stability testing of the memory; testing the first set of overclocked memory settings for performance of the memory and outputting a performance indication; testing the first set of overclocked memory settings for stability of the memory and outputting a stability indication for the first set of overclocked memory settings; selecting at least a second set of overclocked memory settings for performance testing and stability testing of the memory, the second set of overclocked memory settings selected based on the stability indication; and selecting one of the first set or the second set of the overclocked memory settings as optimized overclocked memory settings for the memory.
13. The method of claim 12, further comprising automatically configuring memory to operate with the optimized overclocked memory settings.
14. The method of claim 12, further comprising outputting the optimized overclocked memory settings in a user interface.
15. The method of claim 14, wherein the user interface includes one or more controls to enable adjustment of the optimized overclocked memory settings.
16. A computing device comprising: a memory controller to: initiate a boot up process of the computing device to automatically select optimized overclocked memory settings for a memory; and during the boot up process of the computing device: test performance of a first set of overclocked memory settings; test stability of the first set of overclocked memory settings; test performance of at least a second set of overclocked memory settings; test stability of the second set of overclocked memory settings; and select one of the first set or the second set of overclocked memory settings as the optimized overclocked memory settings.
17. The computing device of claim 16, wherein the memory controller is further configured to automatically configure the memory to operate with the optimized overclocked memory settings.
18. The computing device of claim 16, wherein the memory controller is further configured to output the optimized overclocked memory settings in a user interface.
19. The computing device of claim 18, wherein the user interface enables adjustment of the

optimized overclocked memory settings.
20. The computing device of claim 16, wherein the memory controller initiates the boot up process responsive to user input to a user interface to select optimized overclocked memory settings.